

Multivariate Data Glyphs: Principles and Practice

II.7

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7.1

Introduction

In the context of data visualization, a glyph is a visual representation of a piece of data where the attributes of a graphical entity are dictated by one or more attributes of a data record. For example, the width and height of a box could be determined by a student's score on the midterm and final exam for a course, while the box's color might indicate the gender of the student. The definition above is rather broad, as it can cover such visual elements as the markers in a scatterplot, the bars of a histogram, or even an entire line plot. However, a narrower definition would not be sufficient to capture the wide range of data visualization techniques that have been developed over the centuries that are termed glyphs.

Glyphs are one class of visualization techniques used for multivariate data. Their major strength, as compared to techniques such as parallel coordinates, scatterplot matrices, and stacked univariate plots, is that patterns involving more than two or three data dimensions can often be more readily perceived. Subsets of dimensions can form composite visual features that analysts can be trained to detect and classify, leading to a richer description of interrecord and intrarecord relationships than can be extracted using other techniques.

However, glyphs do have their limitations. They are generally restricted in terms of how accurately they can convey data due to their size and the limits of our visual perception system to measure different graphical attributes. There are also constraints on the number of data records that can be effectively visualized with glyphs; excessive data set size can result in significant occlusion or the need to reduce the size of each glyph, both of which make the detection of patterns difficult, if not impossible. Thus glyphs are primarily suitable for qualitative analysis of modest-sized data sets.

This paper describes the process of glyph generation – the mapping of data attributes to graphical attributes – and presents some of the perceptual issues that can differentiate effective from ineffective glyphs. Several important issues in the use of glyphs for communicating information and facilitating analysis are also discussed, including dimension order and glyph layout. Finally, some ideas for future directions for research on visualization using glyphs are presented.

7.2

Data

Glyphs are commonly used to visualize multivariate data sets. *Multivariate data*, also called multidimensional or n -dimensional data, consist of some number of items or records, n , each of which is defined by a d -vector of values. Such data can be viewed as a dxn matrix, where each row represents a data record and each column represents an observation, variable, or dimension. For the purpose of this paper, we will assume a data item is a vector of scalar numeric values. Categorical and other nonnumeric values can also be visualized using glyphs, though often only after conversion to numeric form (Rosario et al., 2004). Nonscalar values can also be incorporated by

linearizing the embedded vectors or tensors. We also assume that a data set consists of one or more of such data items/records and that for each position in the vector we can calculate a minimum and maximum value. This allows us to normalize the data to facilitate mapping to graphical attributes.

Variables/dimensions can be independent or dependent, which might imply that some ordering or grouping of dimensions could be beneficial. They can be of homogeneous type, such as a set of exam grades, or of mixed/heterogeneous types, such as most census data. This might suggest the use of a consistent mapping (e.g., all dimensions map to line lengths) or separation based on type so that each distinct group of related dimensions might control one type of mapping.

Mappings

7.3

Many authors have developed lists of graphical attributes to which data values can be mapped (Cleveland and McGill, 1984; Cleveland, 1993; Bertin, 1981). These include position (1-, 2-, or 3-D), size (length, area, or volume), shape, orientation, material (hue, saturation, intensity, texture, or opacity), line style (width, dashes, or tapers), and dynamics (speed of motion, direction of motion, rate of flashing).

Using these attributes, a wide range of possible mappings for data glyphs are possible. Mappings can be classified as follows:

- One-to-one mappings, where each data attribute maps to a distinct and different graphical attribute;
- One-to-many mappings, where redundant mappings are used to improve the accuracy and ease at which a user can interpret data values; and
- Many-to-one mappings, where several or all data attributes map to a common type of graphical attribute, separated in space, orientation, or other transformation.

One-to-one mappings are often designed in such a way as to take advantage of the user's domain knowledge, using intuitive pairings of data to graphical attribute to ease the learning process. Examples include mapping color to temperature and flow direction to line orientation. Redundant mappings can be useful in situations where the number of data dimensions is low and the desire is to reduce the possibility of misinterpretation. For example, one might map population to both size and color to ease analysis for color-impaired users and facilitate comparison of two populations with similar values. Many-to-one mappings are best used in situations where it is important to not only compare values of the same dimension for separate records, but also compare different dimensions for the same record. For example, mapping each dimension to the height of a vertical bar facilitates both intrarecord and inter-record comparison. In this paper, we focus primarily on one-to-one and many-to-one mappings, although many of the principles discussed can be applied to other mappings.

Examples of Existing Glyphs

The following list (from Ward, 2002) contains a subset of glyphs that have been described in the literature or are in common use. Some are customized to a particular application, such as visualizing fluid flow, while others are more general purpose. In a later section we examine many of these mappings and try to identify some of their strengths and weaknesses.

- Profiles (du Toit, 1986): height and color of bars (Fig. 7.1a).
- Stars (Siegel et al., 1972): length of evenly spaced rays emanating from center (Fig. 7.1b).
- Anderson/metroglyphs (Anderson, 1957; Gnanadesikan, 1977): length of rays (Fig. 7.1b).
- Stick figures (Pickett and Grinstein, 1988): length, angle, color of limbs (Fig. 7.1c).
- Trees (Kleiner and Hartigan, 1981): length, thickness, angles of branches; branch structure derived from analyzing relations between dimensions (Fig. 7.1c).
- Autoglyph (Beddow, 1990): color of boxes (Fig. 7.1d).
- Boxes (Hartigan, 1975): height, width, depth of first box; height of successive boxes (Fig. 7.1d).
- Hedgehogs (Klassen and Harrington, 1991): spikes on a vector field, with variation in orientation, thickness, and taper.
- Faces (Chernoff, 1973): size and position of eyes, nose, mouth; curvature of mouth; angle of eyebrows (Fig. 7.1e).
- Arrows (Wittenbrink et al., 1996): length, width, taper, and color of base and head (Fig. 7.1f).
- Polygons (Schroeder et al., 1991): conveying local deformation in a vector field via orientation and shape changes.
- Dashtubes (Fuhrmann and Groller, 1998): texture and opacity to convey vector field data.
- Weathervanes (Friedman et al., 1972): level in bulb, length of flags (Fig. 7.1f).
- Circular profiles (Mezzich and Worthington, 1978): distance from center to vertices at equal angles.
- Color glyphs (Levkowitz, 1991): colored lines across a box.
- Bugs (Chuah and Eick, 1998): wing shapes controlled by time series; length of head spikes (antennae); size and color of tail; size of body markings.
- Wheels (Chuah and Eick, 1998): time wheels create ring of time series plots, value controls distance from base ring; 3-D wheel maps time to height, variable value to radius.
- Boids (Kerlick, 1990): shape and orientation of primitives moving through a time-varying field.
- Procedural shapes (Rohrer et al., 1998; Ebert et al., 1999): blobby objects controlled by up to 14 dimensions.
- Glyphmaker (Ribarsky et al., 1994): user-controlled mappings.
- Icon Modeling Language (Post et al., 1995): attributes of a 2-D contour and the parameters that extrude it to 3-D and further transform/deform it.

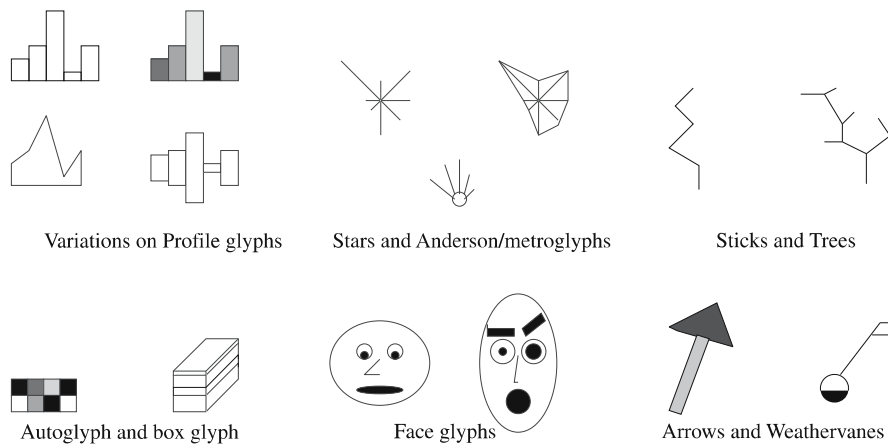


Figure 7.1. Examples of multivariate glyphs (from Ward, 2002)

The list above is ample evidence that a significant number of possible mappings exist, many of which have yet to be proposed or evaluated. The question then becomes determining which mapping will best suit the purpose of the task, the characteristics of the data, and the knowledge and perceptual abilities of the user. These issues are described in the sections below.

Biases in Glyph Mappings

7.5

One of the most common criticisms of data glyphs is that there is an implicit bias in most mappings, i.e., some attributes or relationships between attributes are easier to perceive than others. For example, in profile or star glyphs, relationships between adjacent dimensions are much easier to measure than those that are more separated, and in Chernoff faces, attributes such as the length of the mouth or nose are perceived more accurately than graphical attributes such as curvature or radius.

In this section I attempt to isolate and categorize some of these biases, using both results from prior studies on graphical perception as well as our own empirical studies. It is clear, however, that much more substantial work is needed in measuring and correcting for these biases when designing and utilizing glyphs in data analysis.

Perception-based bias Certain graphical attributes are easier to measure and compare visually than others. For example, Cleveland (1993) reports on experiments that show length along a common axis can be gauged more accurately than, say, angle, orientation, size, or color. Figure 7.2 shows the same data with three different mappings. Relationships are easier to see with the profile glyphs (length on a common base), followed by the star glyphs (length with different orientations). The pie glyph fares the worst, as the user is required to compare angles. Thus in mappings that are not many-to-one (i.e., those that employ a mixture of graphi-

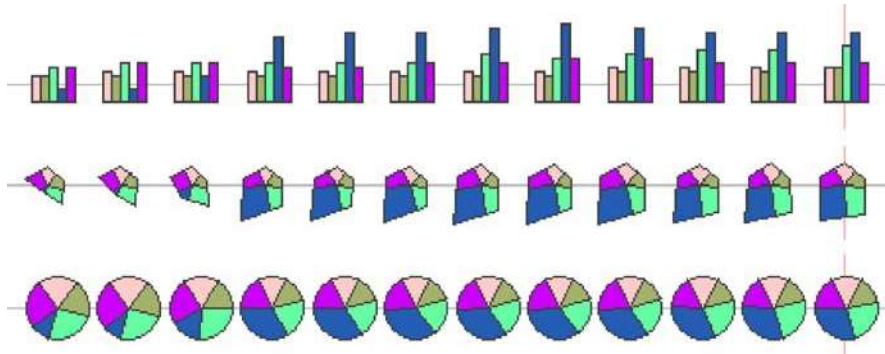


Figure 7.2. Profile, glyphs, and pie glyphs of a subset of data regarding five economic indicators, as generated with SpiralGlyphics (Ward and Lipchak, 2000). Features within and between glyphs are generally easier to compare with profile glyphs

cal attributes), there is an inherent difference in our ability to extract values from different data dimensions.

Proximity-based bias In most, if not all, glyphs, relationships between data dimensions mapped to adjacent features in a glyph are easier to perceive and remember than those mapped to nonadjacent features. To the best of my knowledge, no one has performed experiments to quantify the degree of this bias, although Chernoff and Rizvi (1975) reported as much as 25 % variance in results by rearranging data mappings within Chernoff faces. It is likely that the amount of bias will depend as well on the type of glyph used, as comparing lengths of bars with a common baseline will be easier than comparing the lengths of rays in a star glyph.

Grouping-based bias Graphical attributes that are not adjacent but may be semantically or perceptually grouped may result in the introduction of bias as well. For example, if we map two variables to the size of the ears in a face, the relationship between those variables may be easier to discern than, say, mapping one to the shape of the eye and the other to the size of the adjacent ear.

7.6

Ordering of Data Dimensions/Variables

Each dimension of a data set will map to a specific graphical attribute. By modifying the order of dimensions while preserving the type of mapping, we will generate alternate “views” of the data. However, barring symmetries, there are $N!$ different dimension orderings, and thus distinct views. An important issue in using glyphs is to ascertain which ordering(s) will be most supportive of the task at hand. In this section, I will present a number of dimension-ordering strategies that can be used to generate views that are more likely to provide more information than random ordering.

Correlation-driven

7.6.1

Many researchers have proposed using correlation and other similarity measures to order dimensions for improved visualization. Bertin's reorderable matrix (Bertin, 1981) showed that by rearranging the rows and columns of a tabular display, groups of related records and dimensions could be exposed. Ankerst et al. (1998) used cross-correlation and a heuristic search algorithm to rearrange dimensions for improved interpretability. Friendly and Kwan (2003) introduced the notion of effect ordering, where an ordering of graphical objects or their attributes would be decided based on the effect or trend that a viewer seeks to expose. In particular, they showed that by ordering the dimensions of a star glyph based on their angles in a biplot (basically each dimension is represented by a line whose angle is controlled by the first two eigenvectors), related dimensions would get grouped together. This is related to the method reported by Borg and Staufienbiel (1992), where they compared traditional snowflake and star glyphs with what they called factorial suns, which display each data point using the dimension orientations generated via the first two eigenvectors rather than uniformly spaced angles. Their experiments showed significant improvement by naive users in interpreting data sets.

Symmetry-driven

7.6.2

Gestalt principles indicate we have a preference for simple shapes, and we are good at seeing and remembering symmetry. In Peng et al. (2004), the shapes of star glyphs resulting from using different dimension orders were evaluated for two attributes: monotonicity (the direction of change is constant) and symmetry (similar ray lengths on opposite sides of the glyph). The ordering that maximized the number of simple and symmetric shapes was chosen as the best. User studies showed a strong preference for visualizations using the ordering optimized in this fashion. We conjecture that simple shapes are easier to recognize and facilitate the detection of minor shape variations; for example, shapes with only a small number of concavities and convexities might require less effort to visually process than shapes with many features. Also, if most shapes are simple, it is much more apparent which records correspond to outliers. More extensive formal evaluations are needed to validate these conjectures, however. See Fig. 7.3 for an example.

Data-driven

7.6.3

Another option is to base the order of the dimensions on the values of a single record (base), using an ascending or descending sorting of the values to specify the global dimension order. This can allow users to see similarities and differences between the base record and all other records. It is especially good for time-series data sets to show the evolution of dimensions and their relationships over time. For example, sorting the exchange rates of ten countries with the USA by their relative values in the first year of the time series exposes a number of interesting trends, anomalies, and periods of relative stability and instability (Fig. 7.4). In fact, the original order is nearly reversed at a point later in the time series (not shown).

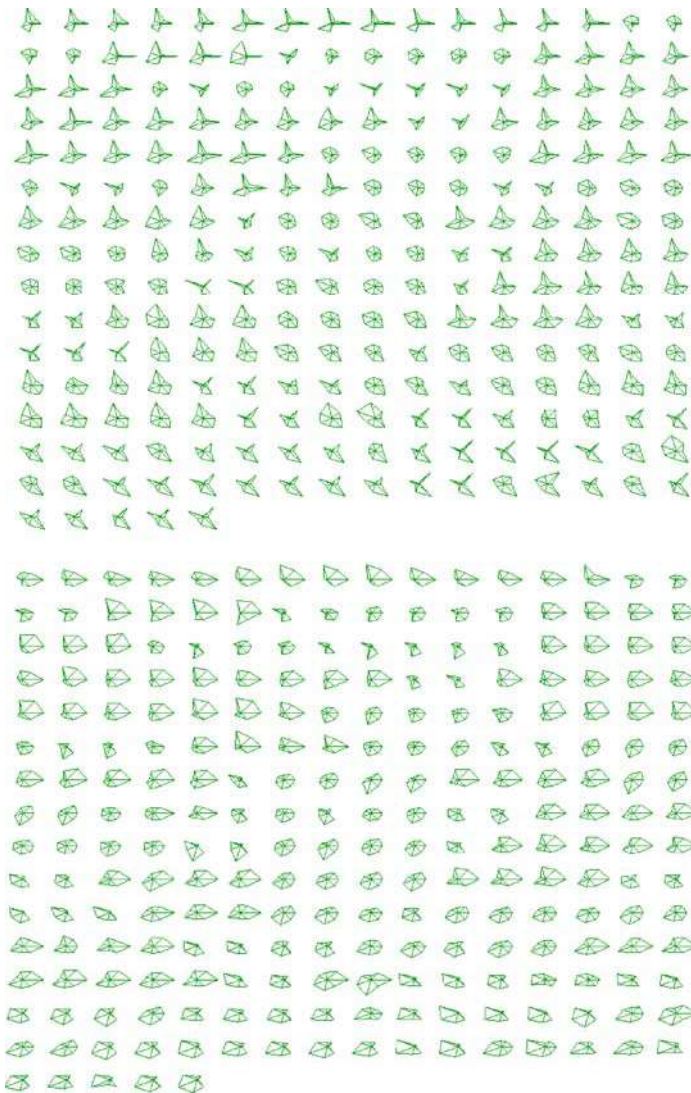


Figure 7.3. Star glyphs for a subset of the cars data set using a random dimension ordering and one based on shape analysis. More simple shapes can be seen in the second version than the first, which we believe can facilitate detection of groups of similar shapes as well as outliers

7.6.4 User-driven

As a final strategy, we can allow users to apply knowledge of the data set to order and group dimensions by many aspects, including derivative relations, semantic similarity, and importance. Derivative relations mean that the user is aware that one or more dimensions may simply be derived through combinations of other dimensions.



Figure 7.4. [This figure also appears in the color insert.] Exchange rate data using the original ordering of dimensions and then ordered by the first data record. Significant features in the ordered version, such as the sudden rise in value of one of the lower currencies during the third year and the progressive alignment of several of the inner currencies, are difficult to detect in the original ordering

These derivative relations might or might not get exposed in correlation-based ordering. Semantic similarities indicate dimensions that have related meanings within the domain; even if the values do not correlate well, users might logically group or order them to help in their analysis task. Finally, some dimensions are likely to have more importance than others for a given task, and thus ordering or assigning such dimensions to more visually prominent features of the glyph (or, in some cases, the features the user is likely to examine first, such as the leftmost bar of a profile glyph) will likely have a positive impact on task performance.

7.7 **Glyph Layout Options**

The position of glyphs can convey many attributes of data, including data values or structure (order, hierarchy), relationships, and derived attributes. In this section I will describe a taxonomy of glyph layout strategies, presented in detail in (Ward, 2002), based on the following considerations:

- Whether the placement will be data driven, e.g., based on two or more data dimensions, or structure driven, such as methods based on an explicit or implicit order or other relationship between data points.
- Whether overlaps between glyphs will be allowed. This can have a significant impact on the size of the data set that can be displayed, the size of the glyphs used, and the interpretability of the resulting images.
- The tradeoff between optimized screen utilization, such as found in space-filling algorithms, versus the use of white space to reinforce distances between data points.
- Whether the glyph positions can be adjusted after initial placement to improve visibility at the cost of distorting the computed position. Overlapping glyphs can be difficult to interpret, but any movement alters the accuracy of the visual depiction. We need to know, for the given domain, what the tradeoffs are between accuracy and clarity.

7.7.1 **Data-driven Placement**

Data-driven glyph placement, as the name implies, assumes a glyph will be positioned based entirely on some or all of the data values associated with the corresponding record. We differentiate two classes of such techniques based on whether the original data values are used directly or whether positions are derived via computations involving these data values. An example of the first would be the positioning of markers in a scatterplot using two dimensions (Fig. 7.5), while an example of the second would be to use PCA to generate the x and y coordinates of the resulting glyph (Fig. 7.6). More complex analysis has also been used in glyph placement. Several researchers (Globus et al., 1991; Helman and Hesselink, 1991) have proposed methods for placing glyphs at critical points within flow fields from fluid dynamics

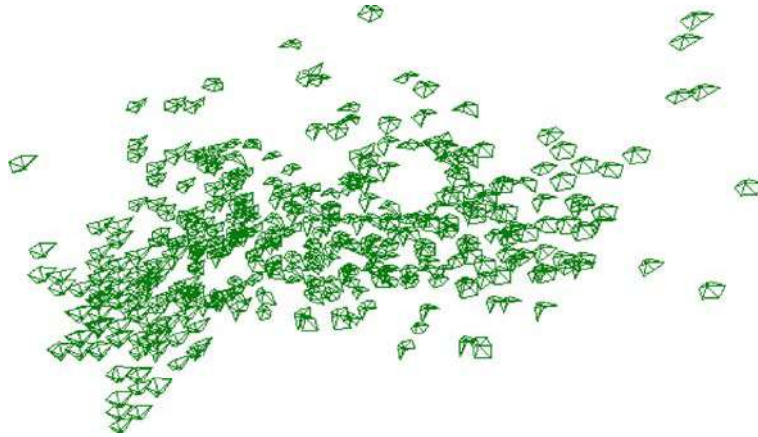


Figure 7.5. Example of positioning glyphs according to two dimensions. In this case, the cars data set displayed with star glyphs using MPG and horsepower to specify position. Groupings of similar shapes and some outliers are visible

simulations. The advantages of using the direct method is that position has an easily interpreted meaning and can act to emphasize or even replace two of the data dimensions. The derived methods can add to the information content of the display and draw the user's attention to implicit relationships between data records.

Data-driven methods almost always result in some overlap between glyphs, which can lead to misinterpretation and undetected patterns. Many methods have been developed to address this problem via distorting the position information. Random jitter is commonly added to positions in plotting, especially for data that take on only a small number of possible values. Other methods use spring- or force-based methods to minimize or eliminate overlaps while also minimizing the displacement of glyphs from their original positions (Keim and Hermann, 1998). Woodruff et al. (1998) proposed a relocation algorithm that attempts to maintain constant density across the display. Since distortion introduces error into the visual presentation, it is best to allow users to control the amount of distortion applied by either setting the maximum displacement for an individual glyph or the average among all glyphs or by using animation to show the movement of glyphs from their original positions to their distorted positions.

Structure-driven Placement

7.7.2

Structure-driven glyph placement assumes the data have some implicit or explicit structural attribute that can be used to control the position. A common type of structure is an ordering relationship, such as in time-series or spatial data. Ordering can also be derived via one or more dimensions (Fig. 7.7). This is different, however, from data-driven placement algorithms in that the data values only define the ordering relationship, which is then used in generating the position. A related structure is a cyclic

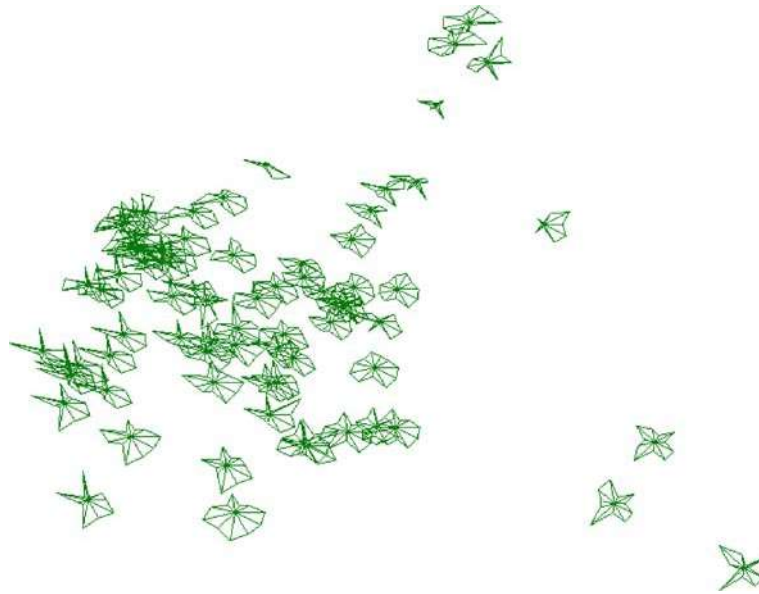


Figure 7.6. Example of positioning glyphs using derived dimensions. In this case, the breakfast cereal data set is shown using the first two principal components to generate the positions. Again, groups of similar shapes are clearly visible

relationship, where on top of the linear ordering is a cycle length, which implies that each glyph is related not only to the adjacent glyphs in the sequence but also the glyphs in the previous and following cycles. Examples of such cyclic placement are shown in Fig. 7.8.

Another type of structure that can be used for positioning is hierarchical or tree-based structures (Ward and Kein, 1997). These may be a fixed attribute of the data (e.g., a computer file system) or computed via, say, a hierarchical clustering algorithm. A wide range of options exist for computing the positions given such a hierarchical structure, as can be seen in the tree-drawing literature (Di Battista et al., 1999). Hierarchically structured glyphs allow easy access both to the raw data as well as aggregation information (Fig. 7.9). Finally, data records might have a network or graph-based structure, such as geospatial data or the web pages associated with an organization. Again, methods from the graph-drawing community can be used to generate the positions for glyphs.

Different structure-driven placement strategies will have different degrees of overlap; a grid layout of ordered data records can assure no overlaps, while tree and graph layouts for dense data sets can result in significant overlap. In cases of overlap, distortion methods are quite common, as structure may be easily preserved and visible even with significant movement of glyphs. Most nonlinear distortion (lens) techniques (Leung and Apperley, 1994) allow the user to view one region of the data space in greater detail than others, shifting the distribution of screen space to provide more

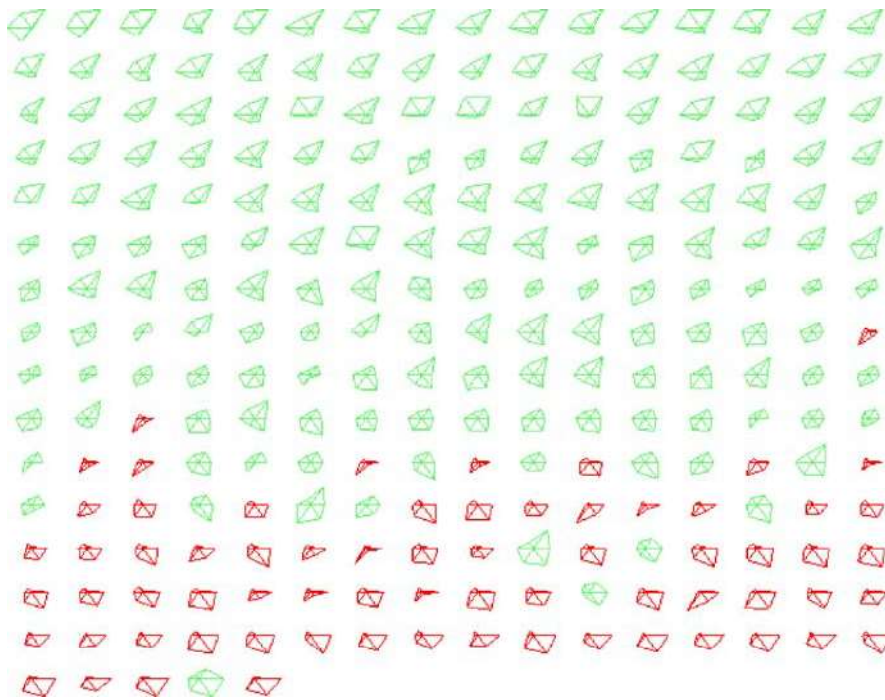


Figure 7.7. Example of ordering glyphs according to one dimension. In this case, the data set is a subset of the cars data, sorted by the MPG variable (*top to bottom* corresponds to low to high MPG). The *highlighted (dark)* glyphs represent four-cylinder cars. A clear grouping of shapes is visible. A few outliers can be seen near the *bottom* of the figure, which represent six-cylinder cars with good MPG

room for the user's focus region(s). This usually enables users to see subsets of the data without the problem of occlusion. Distortion can also be used to enhance separation of subsets of data into groups. Thus in an order-based layout, gaps between adjacent glyphs can be set proportional to a similarity metric. In a sense, this can be seen as a combination of structure and data-driven methods (Fig. 7.10).

Evaluation

7.8

Evaluation of the effectiveness of glyphs for information presentation and analysis can be performed in a number of different ways. In this section, I describe several such assessment processes, including:

- Evaluation based on ranking of human perceptual abilities for different graphical attributes;
- Evaluation based on the speed and accuracy of users performing specific tasks;
- Evaluation based on ease of detection of data features in the presence of occlusion and clutter; and

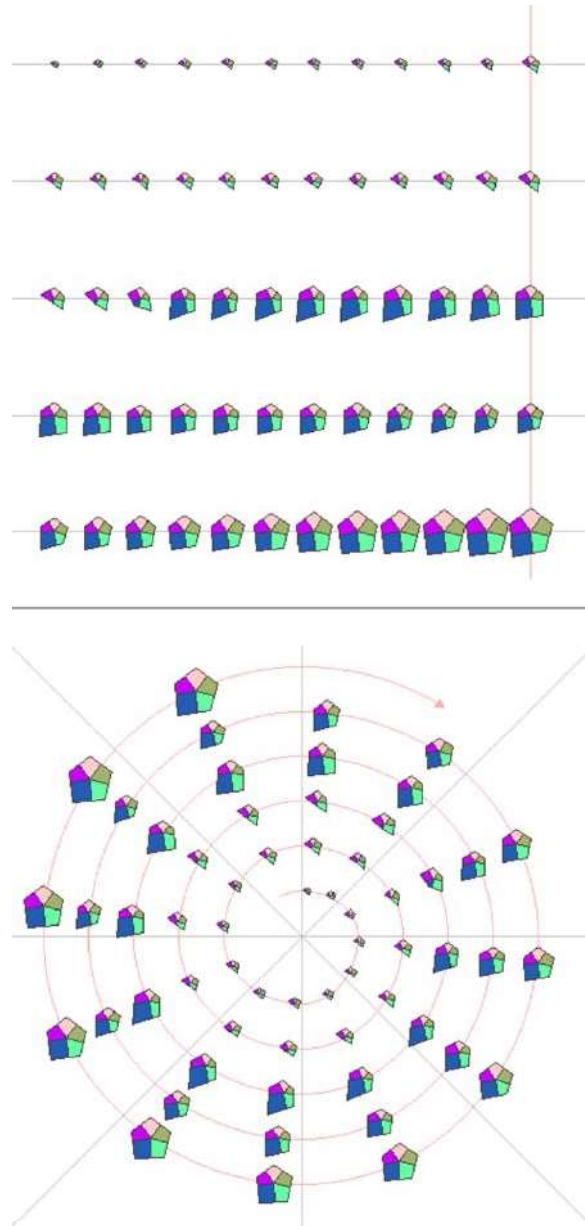


Figure 7.8. Examples of cyclic data glyph layouts, as generated by SpiralGlyphics (Lipchak and Ward, 1997; Ward and Lipchak, 2000). The data consist of five economic indicators over 5 years. In the first layout, each *row* constitutes a cycle, while in the second, each ring of the spiral is one cycle. While both allow the user to see both intracycle and intercycle variations in the data, the spiral more easily allows comparison of the end of one cycle and the beginning of the next. In addition, space appears to be more effectively utilized in the spiral layout for long cycles

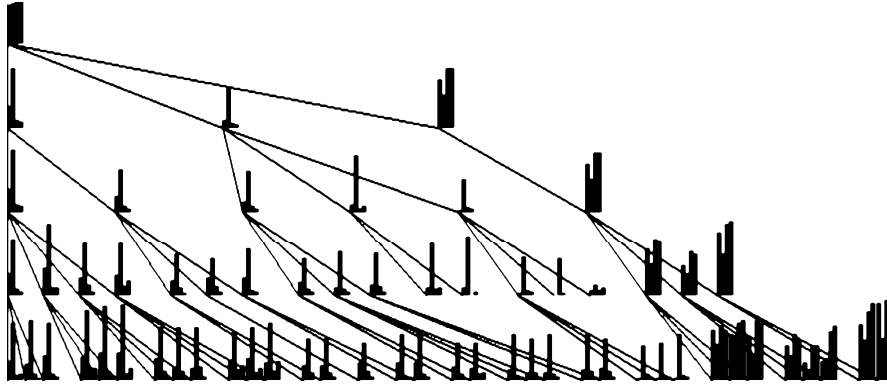


Figure 7.9. Profile glyphs of a hierarchically clustered subset of the Iris data. Nonterminal nodes are computed as the average values of their descendents. The clustering algorithm appears to have done a reasonable job, though a few outliers exist, such as the cluster associated with the fifth node in the third row

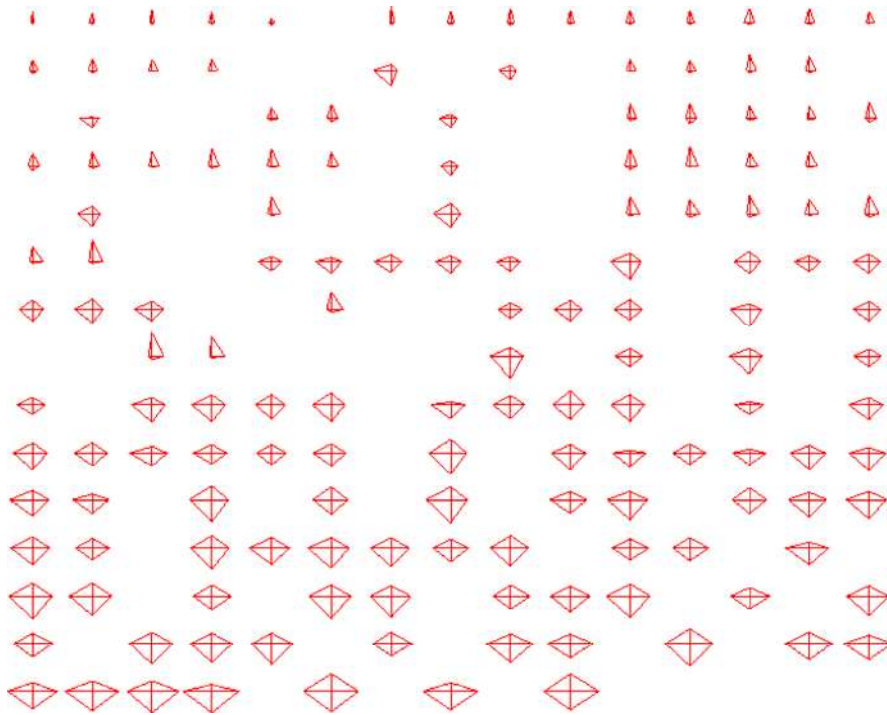


Figure 7.10. Star glyphs of Iris data set, ordered by one dimension and positioned with horizontal spacing proportional to the distance between adjacent data points. This nonoverlapping layout makes it easy to identify both clusters and large gaps in the N-D distance where the values for the ordering dimension are similar to each other

- Evaluation based on the scalability of techniques in terms of number of records and dimensions.

A large number of evaluation studies on glyphs and other forms of multivariate data analysis have been carried out over the years. Some of these have been rather ad hoc, or based just on the opinions and observations of the authors, while others have involved detailed and carefully orchestrated user studies.

Cluff et al. (1991) evaluated and categorized 12 methods of multivariate data presentation, including several forms of glyphs. Their evaluation criteria fell into three groups: objectives, information level and dimension capacity, and global criteria. Under the objectives group they considered accuracy, simplicity, clarity, appearance, and design. In the information level and dimensional capacity criterion, visualizations are classified as to what extent they retain the level of information present in the data. At the lowest level (elementary) is the ability to convey the individual data values, while at the intermediate level, relationships among subsets of the data can be seen. The highest level (overall) provides linkages between multiple relationships and allows users to understand the data sufficiently to solve real tasks. The global criteria group includes flexibility, interpretability, visual impact, time to mastery, and computational tractability. For each visualization method, each of these criteria was classified (subjectively by the authors) as 1 = does not sufficiently meet objective, 2 = meets objective satisfactorily, and 3 = meets objective in an excellent manner. While the results of the classifications lack statistical significance due to the small sample size, the categorization of evaluation criteria was far more extensive than in most other studies.

Lee et al. (2003) analyzed the effectiveness of two glyph techniques (Chernoff faces and star glyphs) and two spatial mappings, where each data record was simply represented by a marker whose position was based on similarity to other data records. Binary data was used, and 32 subjects were asked a range of questions regarding relationships between records (both local and global). Results showed that the subjects answered many of the questions more quickly and accurately, and with more confidence, using the spatial mappings. This confirmed the hypothesis of many researchers, which is that glyph interpretation can be quite slow for tasks that involve a significant scanning and comparison. However, questions regarding the values of particular data features could not readily be answered with the spatial mappings. The implication is that, given a known set of questions, it may be possible to assign positions of simple points to facilitate a task. For general tasks, however, a combination involving positioning of glyphs based on data relationships, as suggested in the glyph layout section of this paper, would likely be most effective.

As mentioned earlier, Borg and Staufienbiel (1992) compared snowflake and star glyphs with factorial suns, with the angles of the lines conveying relationships between dimensions. In their experiment, they used the classification of 44 prototypical psychiatric patients across 17 attributes into 4 categories, as determined by 11 experts. Then, using each of the 3 glyph types, they generated drawings of each of the 44 cases. Thirty beginning psychology students were asked to group the drawings into 4 categories based on shape similarity. They then studied the frequency with which drawings of the same category (according to the experts) were grouped together by the

students. The results showed a significantly greater success rate with factorial suns as compared to snowflakes and star glyphs.

Several evaluations of Chernoff faces have been reported since their introduction. One interesting set of experiments was reported in Morris et al. (1999), who focused on studying the effectiveness and preattentiveness of different facial features. Subjects were shown images containing varying numbers of faces, and they were asked to determine if a face with a designated feature existed. The amount of time required to complete each task was measured and analyzed. Not surprisingly, the amount of time needed was proportional to the number of glyphs on the screen. However, the authors determined that it is not likely that preattentive processing was involved, as tests done with short duration, even with small numbers of glyphs, yielded poor results. Their conclusion was that, because glyph analysis with Chernoff faces was being done sequentially, they are unlikely to provide any advantage over other types of multivariate glyphs.

A study that placed Chernoff faces ahead of several other glyph types was reported by Wilkinson (1982). In this study, subjects were asked to sort sets of glyphs from most similar to least similar. The glyphs used were Chernoff faces (Chernoff, 1973), Blobs (Andrews, 1972), castles (Kleiner and Hartigan, 1981), and stars (Siegel et al., 1972). The results were that faces produced results with the best goodness of fit to the real distances, followed by stars, castles, and blobs. The author felt that the memorability of the faces helped users to better perform this type of task.

Summary

7.9

Glyphs are a popular, but insufficiently studied, class of techniques for the visualization of data. In this paper, we've discussed the process and issues of glyph formation and layout, including the identification of problems of bias due to perceptual limitations and dimension ordering. We also presented techniques for evaluating the effectiveness of glyphs as a visualization method and some results obtained from evaluation.

Many avenues exist for future development and application of glyphs for data and information visualization. There is a continuing need for glyph designs that minimize bias while maximizing the accuracy of communicating data values. While the majority of recent designs have been tailored to particular domains and tasks, we believe there is still room for work on general-purpose glyph designs. Scalability is also a big issue, as most glyph methods in use are limited either by the number of data records or data dimensions that can be easily accommodated. Given the growth in the size and dimensionality of common data sets, novel mechanisms are needed to enable users to explore larger and larger amounts of data. Work on aggregation glyphs (Yang et al., 2002) or other multiresolution strategies may be the key to the problem of scale. Finally, rigorous evaluation is essential to help identify the strengths and weaknesses of each proposed glyph in terms of user perception, analysis tasks, and data characteristics. Most efforts at evaluation to date have been ad hoc or very limited in scope.

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